ChatGPT

SilicaFlux Production Test & Qualification Plan

Executive Summary

SilicaFlux-V (the validation prototype) and the full SilicaFlux SoC will require a comprehensive *design-for-test (DFT)* and manufacturing validation strategy. This plan covers all phases from RTL design (DFT insertion, scan and BIST) through wafer probing and ATE testing, to reliability qualification and production release. Key elements include full-scan chains for core logic, IEEE 1149.1 JTAG boundary-scan for I/Os, on-chip BIST for each SRAM block (using March test algorithms【19†L114-L122】【58†L208-L214】), and well-defined test programs on automated testers. We will bin devices by speed grade and subject them to accelerated stress tests (e.g. JESD22-A108 HTOL, A110 HAST) and ESD/latch-up screening【39†L73-L82】. The plan defines supplier roles (foundry, packaging, test house) and success criteria at each stage (e.g. fault coverage targets, yield thresholds).

The deliverables include a DFT/tapeout readiness checklist, a schedule (engineering sample → pilot → production), a mermaid flowchart of the test-program architecture, tables linking RTL blocks to DFT hooks/test vectors, recommended ATE and probe-card specifications, and a failure-analysis escalation matrix. This ensures that SilicaFlux chips can be reliably tested, debugged and qualified for volume manufacturing.

1. Design-for-Test Architecture

- **Scan Chains:** We will insert full-scan on all sequential elements (flip-flops, registers) in each logic block. Scan chains maximize test controllability/observability【3†L136-L144】【24†L7-L10】. Clock-domain crossings will use isolated chain heads/tails or reset converters. Scan chains should be balanced (similar length) across blocks to optimize ATE usage. High-speed clocks should have slow “shift” paths for scan to avoid timing issues (as in Fig.1 of [11]).
- **ATPG and Test Points:** After scan insertion, we use ATPG tools (e.g. Synopsys TetraMax, Mentor Tessent) to generate stuck-at and transition-fault vectors【24†L7-L10】. Poorly testable nodes (long paths, dense logic) may need inserted test points (half-latches) to improve coverage. We target >95% structural fault coverage; all remaining faults are to be caught by BIST or functional tests【24†L7-L10】【24†L25-L29】.
- **BIST:** Each SRAM (L1 instruction/data caches, L2 banks, scratchpad, register file, etc.) will include an on-chip memory BIST engine. A standard March algorithm (e.g. March C−) will run on each memory to detect stuck-at, coupling and address-decoder faults【19†L114-L122】. Embedding BIST drastically reduces dependence on external test vectors and speeds up memory verification【58†L208-L214】.
- **Boundary-Scan/JTAG:** A 5-pin JTAG TAP (TCK, TMS, TDI, TDO, optional TRST) will be exposed on package pins. All bidirectional I/O pads (GPIO, UART pins, DDR interface pins, etc.) will include IEEE-1149.1 boundary-scan cells. This allows board-level interconnect testing and some at-speed access to on-chip registers. IEEE-1149.1 is the industry-standard test access port, providing far greater test coverage than ad-hoc methods【55†L160-L167】【23†L139-L148】. We may adopt IEEE-1149.7 (compact TAP) to reduce pin count if needed.
- **DFT Insertion:** DFT logic (scan, BIST controllers, JTAG TAP) will be inserted early in RTL phase【3†L136-L144】. We will use a DFT insertion toolchain (Open-source Yosys/ABC for initial flows,

then commercial Synopsys/Mentor tools for signoff). Care must be taken to back-annotate scan timing into STA and to constrain scan chains (clock gating, asynchronous resets) properly.

Citations: Proper DFT (scan/BIST/jTAG) is embedded during design to enable high fault coverage【3†L136-L144】【58†L208-L214】. In practice, ATPG vectors are applied via the chip's scan chains or BIST【24†L7-L10】. IEEE 1149.1 (JTAG) is widely used as the de-facto test access port because it "enables a much greater test coverage than any other test technology"【55†L160-L167】.

2. Scan, ATPG, and Compression Strategy

- **Scan Architecture:** We will organize scan chains by block (e.g. one per CPU core, one per cache bank, one per datapath). Within each chain, a mux selects between functional input and scan-in. The chain outputs connect to scan-out pins or to the next chain. At least one internal scan-in/out pin per core and memory cluster will be brought out to the TAP (via the IEEE-1500 "test bus" or embedding in JTAG).
- **ATPG Flow:** Post-DFT RTL/netlist will be fed to ATPG. We will target stuck-at and transition-fault models in early pilots. Fault coverage reports guide insertion of additional test points if needed. We expect 95–98% coverage of classic faults; unreachable faults (e.g. from trim cells) may be exempt if justified. The ATPG patterns become the core of the ATE test program.
- **Test Compression:** To reduce test data volume and time, we plan to use scan-chain reconfiguration or standard LFSR-based compression (if supported by the test tools/ATE). For example, multiple scan chains can be fed in parallel from a Linear Feedback Shift Register, as is common in modern DFT. This can cut vector count by ~10× in some cases (the exact number depends on chain count and coupling). In short, test compression is required to avoid impractically long test times on ATE【11†L339-L347】.
- **Power-Aware Testing:** High-speed scan shift toggles many flops simultaneously, which can cause IR drop and local hot spots【11†L339-L347】. We will set conservative scan clock rates (tens of MHz) and possibly implement "safe mode" shift (e.g. divide-by-4) to limit switching current. ATPG tools will be run in "power-aware" mode to control toggling during test pattern creation【24†L25-L29】.
- **Toolchain:** An example flow is: RTL w/ DFT → DFT synthesis (insert scan/BIST/JTAG) → netlist ATPG (TetraMax/Tessent) → generate .vec and .wgl (waveform) files → ATE program builder. We will establish a CI process to run ATPG and collect coverage after each major design change.

Citation: Integrating ATPG early and tailoring DFT improves test coverage and reduces test time and cost【24†L25-L29】. In scan testing, tester time is significant – one estimate is that testing can consume up to 50% of the chip's cost due to long shift-in/shift-out cycles【11†L339-L347】. Thus, compression and scan optimization are critical.

3. Memory BIST (MBIST)

- **Per-Memory BIST Engines:** For each SRAM block (L1 instruction cache, L1 data cache, L2 cache banks, scratchpad RAMs, etc.), we will instantiate a built-in self-test controller. The BIST will apply a March-like algorithm (e.g. March C− or March B+), which is known to cover stuck-at, address decoder, and coupling faults【19†L114-L122】.
- **March Patterns:** A typical March C− sequence might be "↑(w0) ↑(r0,w1) ↓(r1,w0,r0)" which writes and reads all 0/1 in ascending/descending addresses. We will include as needed some extra patterns (e.g. checkerboard, walking 1's/0's) to catch pattern-sensitive faults.
- **BIST Architecture:** The MBIST logic will include address generation, pattern generator, data comparator and an output MISR (signature). During test mode (entered via a JTAG command or a

dedicated BIST start pin), the MBIST iterates through the memory rows autonomously. A final fail flag or signature is captured and read out. This frees the ATE from needing to stream millions of memory vectors.

- **Integration:** MBIST controllers are typically attached at the RTL or GDS level. We will either use a memory compiler that includes MBIST (if available) or hand-assemble an open MBIST IP (Marinescu BIST algorithm) into each block. Control registers for starting MBIST and reading results will be exposed via the system bus.

Citation: SRAM BIST (using March tests) is standard for production test of on-chip memories【19†L114-L122】. Embedding BIST "makes faster, less-expensive IC manufacturing tests" by offloading test pattern generation to the chip【58†L208-L214】.

4. JTAG & Boundary-Scan

- **TAP Logic:** We include an IEEE-1149.1 TAP controller and instruction register. JTAG access allows shifting data into boundary-scan cells and optional internal scan-chains. Common instructions (EXTEST, SAMPLE, BYPASS, etc.) will be supported.
- **Boundary-Scan Cells:** All external I/O (pins and pads) get 1-bit boundary scan cells. In test mode, these cells isolate the core logic from the pins, allowing the TAP to drive or sample each pin. This is invaluable for board-level interconnect test and verifying packaged chip I/O without a CPU running.
- **Usage:** During wafer sort and final test, we will use JTAG to load test patterns (for example, driving DDR data lines in a loopback mode or toggling GPIOs). Also, chip bring-up firmware will use JTAG for debug/read internal signals.
- **IEEE-1687 (IJTAG):** If we embed any on-chip instruments (e.g. performance monitors) we may use IEEE-1687 (IJTAG) to access them via the JTAG TAP, but this is optional.
- **Security:** After test, JTAG can be disabled or locked to prevent tampering in-field. Test escape paths will be closed for final production if required by security.

Citation: Boundary-scan (JTAG) "enables a serial data stream to access the state of nodes within the circuit" without physical probing【23†L63-L72】. Its adoption by IEEE as 1149.1 underscores its importance. In fact, the standard is "widely used" because it achieves far higher coverage for board and chip tests than alternatives【55†L160-L167】.

5. Wafer-Probe Strategy

- **At-Wafer Test:** Before dicing, each die will be probed with a *wafer probe station* to weed out gross defects. We will design dedicated "probe pads" (usually power/ground pads and a few I/Os) for parametric and scan tests. If the design is very pin-limited, we may use probe-only pads for core power rails and JTAG signals.
- **Probe Card:** A reusable probe card (spring-pin pogo or MEMS) will be fabricated for the target wafer size. Early engineering samples can use a generic 68-pin spring-probe card if gate count is small, but for larger designs we expect a 150–300 pin probe card. The card must route at least: all power rails, ground, JTAG pins, clock, a handful of data pins, and any high-speed I/O. Multi-site probing (testing several die in parallel) can shorten test time but is complex.
- **Wafer Sort Tests:** On the ATE wafer tester, we will run:
- **DC Parametric Tests:** Check VDD/VSS short, open, leakage currents (IDDQ), I/O voltage thresholds.
- **Clock Presence:** Verify PLL/clock chip wakes up by toggling signals.
- **Functional/Scan Tests:** Run a subset of scan patterns to check basic logic; exercise MBIST to test caches.

- **Analog Blocks:** If any analog (ADCs, refs) exist, perform basic functional checks via test pattern DAC inputs or loopback.
- **Throughput:** At production speeds, the tester may handle 50–200 pads per second. Extensive test patterns (millions of vectors) are slower, but wafer sort usually uses a trimmed test set focusing on catch-critical-faults. Any chip that fails wafer sort is marked bad to avoid packaging cost.
- **Data Collection:** The wafer sort yields first-pass metrics: gross yield, distribution of failures (e.g. % failing memory, logic). This informs whether the design or process needs correction.

  Remark: Wafer probe test time is often far less than final test time, because we tend to run only a few critical vectors plus MBIST. However, initial engineering wafers might be exhaustively tested to debug failures.

6. Final Package and PCB Bring-up

- **Package Selection:** Early engineering samples can use an open 144-pin QFN or BGA package (exposed pad for heat). QFN is simple and thermally good【34†L102-L105】. For the full SilicaFlux, a fine-pitch BGA (e.g. 0.8 mm pitch, 256–512 balls) is likely needed to accommodate I/Os and power. BGA ensures low inductance and high I/O count.
- **PCB Reference Design:** We will design a simple board for bring-up, including:
- Regulated power rails (with sequenced power-up),
- Decoupling and power dissipating (heatsink on exposed pad),
- High-speed connectors (e.g. DDR memory socket if external DRAM, debug header, JTAG header),
- Possibly FPGA (to emulate off-chip components).
- **Power-Up Sequencing:** If the chip has multiple power domains (core, I/O, memory), the PCB must respect sequencing. A typical safe sequence is to bring up I/O and auxiliary rails before core rails. We will include supervision ICs to ensure stable rails before releasing resets.
- **Bring-Up Tests:** Upon power-up, we first use JTAG to check TAP ID and boundary cells. Then we load a simple “bootloader” via JTAG or SPI, which toggles some I/Os to verify connectivity. Early firmware will turn on MBIST and report results via UART or JTAG. Each major block (CPU, cache, interrupts, peripherals) will have self-check routines in firmware.
- **Thermal/Power Analysis:** We will analyze worst-case power draw (e.g. all cores + vector units active) and ensure the package/PCB can dissipate the heat. Thermal imaging during bring-up will verify hotspots. If the package is QFN, a copper ground pad and vias under the die improve heat removal【34†L130-L138】.
- **Safety:** Power ramps will be slow (soft-start regulators) to avoid inrush. Firmware will start in a safe mode (clock gate most blocks, JTAG debug open) until core functionality is proven.

  Citation: Quad-Flat No-Lead (QFN) packages “offer a small form factor and efficient thermal performance”【34†L102-L105】. For high-pin designs, a BGA with exposed paddle is recommended. Proper PCB decoupling and thermal pads are critical.

7. Functional Test Vectors and Coverage

- **Functional vs. Structural Test:** Structural tests (scan/ATPG) cover manufacturing defects. Functional test ensures the chip actually works as intended. We will write software test suites (runnable on the device) that execute known instruction sequences and use peripherals, to catch logic or microarchitectural bugs.
- **Vector Development:** Key functional tests include: CPU sanity (register moves, branch instructions, L1 cache fills/evicts), vector/tensor unit operations, interrupt handling, DMA

transfers, and I/O loopbacks. Complex scenarios (e.g. cache thrash, context switch) will be scripted. These vectors are often validated on FPGA/emulation first.

- **Automation:** Final test vectors will run on an ATE or high-speed logic board emulating inputs (clock, reset, memory). We can use tools like Synopsys Verdi or Synopsys VC Verification IP to create expected output patterns.
- **Coverage Targets:** We set quantitative goals, e.g. 100% instruction coverage (each ISA instruction executed), 100% register writes tested, and all bus interfaces (AMBA/AXI) exercised. For scan/ATPG, 95% stuck-at and 90% transition coverage is a typical acceptance target.
- **Test Time:** The total vector count (structural + functional) could be millions. However, compression (see above) reduces vector memory. Test time must be kept reasonable (for example, <10 s per device on a volume ATE slot) to maintain cost targets【11†L339-L347】.

8. Speed Grading and Burn-In

- **Speed Binning:** After functional correctness is verified, each chip is speed-tested at maximum rated voltage and temperature extremes. We will define speed grades (e.g. 0.9GHz, 1.0GHz, 1.1GHz) based on the highest clock frequency at which it passes. ATE timing margin test sweeps the clock or toggles its highest-speed loop (PLL + CPU core) to find max frequency. Chips are labeled (or binned) by speed grade.
- **Thermal Binning:** Similarly, we may classify chips by performance at high and low temperature. For example, “Temp A” grade passes at 85°C ambient, while “Temp B” passes up to 105°C. This is important for automotive/performance markets.
- **Burn-In:** A portion of chips (e.g. 100% of automotive-grade or 10% of commercial) will undergo thermal cycling and burn-in stress (e.g. 125°C for 48–168 hours under clock at nominal voltage). This accelerates latent defect failures. If the design is for high-reliability, then we adopt JEDEC burn-in standards (e.g. JESD-22-A108, a 168-hour bias test)【39†L73-L82】.
- **Screening:** Burn-in helps in qualifying samples; parts that fail during burn-in are discarded. For less critical runs, “high-temp operating life” (HTOL) of 100–500 hours may suffice.

Citation: Standard qualification tests include *Temperature-Humidity-Bias Life* (JEDEC JESD22-A101) and *HTOL* (JESD22-A108) to stress package reliability【39†L73-L82】. These accelerated tests are essential for ensuring long-term reliability.

9. Reliability Qualification and Yield Monitoring

- **Accelerated Stress Tests:** To qualify the product, we will perform:
- **HTOL:** Operate chips at elevated temperature (e.g. 125°C) and voltage for 1000 hours (or shorter pilot tests) to catch early life failures【39†L73-L82】.
- **HAST:** For moisture-sensitive technologies, a Highly-Accelerated Stress Test (JESD22-A110) subjects devices to high humidity and bias.
- **Temperature Cycling:** Thermal cycling (−55°C to +125°C, JESD22-A104) to reveal solder or package fatigue.
- **ESD/IO:** Test Human-Body-Model and Charged-Device-Model ESD per JEDEC/ESDA standards (e.g. JESD22-A114, A115) on all I/O pins.
- **Latch-up:** Perform EIA/JESD78 latch-up test by stressing with ± voltage/current to ensure no parasitic thyristor triggers.
- **Reliability Goals:** We aim for automotive-grade quality metrics (e.g. <10 FIT rate per million hours). Qualification samples (typically 77 units) must pass without failures as per JEDEC AEC-Q100 or relevant spec.

- **Yield Learning:** At each production stage, yield will be logged. Low yield (<80%) or systematic failures (>1% devices failing same test) trigger a root-cause analysis. Statistical process control (SPC) charts will be used to spot trends.
- **Sampling Plan:** We will follow MIL-STD-105E sampling or industry norms. For example, from each wafer lot, test ~30 devices for parametric and endurance stress; acceptance based on zero or one failure allowed for small samples. For production, continuous sampling (e.g. 0.1%) will monitor quality.

Citation: JESD-22 defines the standard stress tests for packaged ICs【39†L73-L82】. For example, A108/HTOL and A110/HAST are JEDEC standards for accelerated life testing. Passing these tests is required for qualification.

10. Test Equipment and Infrastructure

- **ATE Requirements:** We expect to use high-end digital ATE for final test. Example configurations:
- **Engineering/Average-Volume:** A Teradyne UltraFLEX or Advantest T5831 tester with ~256–512 digital channels, 100–500 MHz pattern rate, mixed-signal options. It should have >4 power supply modules (for separate core, I/O, PLL rails) and analog measurement units for IDDQ or V-I curves.
- **High-Volume:** A multi-site parallel tester (e.g. Advantest T2000 series) with hundreds of sites to test many chips simultaneously. This requires large capital but amortizes on volumes.
- **Probe Station:** Automated wafer prober with thermal chuck (0–125°C) and precision stage. Probe needles (<100 µm tip) or a MEMS probe card.
- **Probe Card Specs:**
- **Pin Count:** Prototype: ~64 pins (JTAG + essential IOs). Production: ~128–256 pins (to support DDR, many GPIOs).
- **Type:** Spring-pin array for early prototypes; hardened MEMS or copper blade cards for volume.
- **Pad Geometry:** Design the chip's probe pads with standardized pads (e.g. 250µm square) and keep them on a 0.65–1.0 mm grid for cards.
- **Burn-In Systems:** Forced-air ovens or flow-through burn-in boards capable of >100 °C and 5–10 V stress across devices (with temperature/voltage monitoring).
- **Failure Analysis Lab:** We will engage a qualified FA lab (or build in-house) with: SEM imaging, FIB cross-section, thermal imaging camera, decapsulation tools, X-ray (for bond inspection), etc.
- **Software/Test Data:** Use industry-standard test programming languages (STIL, VCD, or python scripts) to generate ATE programs. Results (fail logs, param data) will feed back into a database (e.g. JIRA or Test Data Systems) for metrics.

Costs: Typical turnkey ATE costs range from €100K–€500K per year for engineering capacity. Probe cards cost €5K–€20K each. Burn-in per oven run (20 boards) is ~€2K. (These are ballpark; actual quotes needed.)

11. Supplier Responsibilities

- **Foundry:**
- *Deliverables:* GDSII wafers, preliminary process characterization. May also provide rough wafer sort (power shorts, IDD) if contracted.
- *Requirements:* Supply the PDK, standard-cell libraries, I/O/PPL macros, and recommended DFT guidelines.
- **Memory Compiler IP:** If we use a commercial SRAM compiler with built-in BIST, the memory vendor must supply BIST controllers and support integration.
- **Packaging House:**

- *Deliverables:* Die attach, wirebonding or flip-chip, mold or lid, assembly. Provide board-out test results (Basic DC/Pin continuity).
- *Requirements:* Ensure package meets thermal and pinout specs. Provide recommended land-pattern and reliability test results (JEDEC moisture sensitivity rating).
- **ATE/Test House:**
- *Deliverables:* Final test programs (patterns, tunings) and test execution for wafer sort and final test. Possibly turnkey test-inserted services.
- *Requirements:* Use our test vectors, maintain temperature/stress conditions, deliver pass/fail reports and log of each UUT. They should also offer partial debug support for first lots.
- **PCB/Board House:** Produce bring-up boards and burn-in boards. Source I/O connectors, regulators, crystals.
- **Test Equipment Vendors (ATE, Burn-In):** Provide timely support and maintenance. Possibly loaner programs for initial engineering.
- **FA Lab:** Perform failure analysis on returned test fails, within weeks of request. Provide reports linking electrical symptoms to physical root causes.

12. Deliverables & Acceptance Criteria

For each phase we define entry/exit criteria:

- **Design Freeze (Pre-silicon):** All RTL DFT features in place; ATPG achieved ≥95% coverage; all regressions passed (CIV, STA, DFT-vectors).
- **Tapeout Readiness:** Successful LVS/DRC on test-structure-inserted layout; formal check of JTAG chain; simulated MBIST pass; redlines done.
- **Engineering Samples:** Wafer probe yields >50% bare-die good; basic scan/BIST tests all pass; CPU runs simple code from ROM; no FATAL fails in parametrics.
- **Pilot Lot:** >80% yield after packaging; all speed-grade bins identified; sample passed burn-in; no catastrophic fails.
- **Production Release:** Consistent yields above target (e.g. 90%); 0 escape failures in 1000 samples; reliability tests show no failures.

These milestones form the checklist we track (see Appendix A timeline).

Appendix A: Gantt Timeline (Mermaid)

The timeline below illustrates a suggested schedule from design completion through production qualification (dates are indicative).

```
gantt
    title SilicaFlux Production Test & Qualification Timeline
    dateFormat  YYYY-MM
    section Design & DFT
    DFT Insertion & Scan Insertion      :active, des1, 2026-11, 6m
    ATPG & BIST Pattern Generation     :des2, after des1, 3m
    DFT Verification (Sim/STA)         :des3, after des2, 2m
    section Tapeout
    Mask Layout & Tapeout              :milestone, 2027-07, 0d
    Fab (MPW run for SilicaFlux-V)     :fab1, 2027-08, 2m
    section Silicon Tests
    Wafer Probe & Sort                 :test1, after fab1, 1m
```

```
    Engineering Sample Board Bring-up  :test2, after test1, 2m
    Pilot Lot Fabrication              :fab2, 2027-12, 2m
    Pilot Wafer Sort & Assembly        :test3, after fab2, 1m
    Pilot Functional & Stress Tests    :qual1, after test3, 3m
    section Production Ramp
    Volume Production Start            :prod1, after qual1, 2028-05, 12m
    Ongoing Reliability Monitoring     :prod2, after prod1, 12m
```

Appendix B: RTL Modules vs. DFT Hooks

RTL Module	DFT Hooks Required	Test Vector Focus
CPU Core (Pipeline & ALU)	Full scan chains (scan-in/out on all regs)	ALU ops (add/sub/mul), branch/jump sequences, register file read/write patterns
Vector/Tensor Unit	Scan chains on vector registers, optional BIST	SIMD MAC operations, reduction/accumulate patterns, data bus transfers
L1 Data Cache (SRAM)	On-chip MBIST (March C variants)	Read/write all-0/all-1, checkerboard and inverse patterns, address toggle (start/end) tests
L1 Inst Cache (SRAM)	On-chip MBIST	Instruction fetch sequences, linear address walks
L2 Cache Banks (SRAM)	On-chip MBIST per bank	Multi-bank simultaneous access, conflict/starvation tests
TLB / MMU	Scan chains	TLB hit/miss patterns, page-table reads
Interconnect / Crossbar	Scan chains	Bus/strobe toggling, idle/busy arbitration scenarios
DRAM Controller	Scan chains	DDR read/write, calibration patterns, data-eye test patterns
Interrupt Controller	Scan chains	Mask/unmask interrupts, irq edge and level tests
GPIO and Peripherals (UART, I2C, SPI, etc.)	Boundary-scan (JTAG) for pins + scan if internal regs	Pin toggling, loopback (e.g. connect TX to RX), SPI transaction sequences
PLL / Clock-Reset Gen	Scan chain (test mode for divider)	Disable PLL, force reference clock, verify output clocks via scan capture
JTAG TAP & Bypass	IEEE-1149.1 TAP logic (inherently DFT)	Basic instruction tests (IDCODE, BYPASS register, boundary-scan shift)

Each entry above corresponds to RTL source files for that block. The DFT hooks indicate insertion of scan or BIST. Test vectors are examples of focused tests to exercise that block's functionality. (In practice, the actual ATPG patterns will be generated automatically based on these points.)

Appendix C: Test-Program Architecture (Mermaid Flowchart)

The flowchart below depicts the end-to-end testing architecture from DFT-inserted design to qualification.

```
flowchart LR
    subgraph Design
        DFT[“Insert Scan/BIST/JTAG”]
        ATPG[“ATPG & Test Vector Generation”]
        DFT --> ATPG
    end
    subgraph Test_Exe
        ATE[“ATE Test Program & Execution”]
        Probe[“Wafer Probe Tests”]
        Final[“Final (Package) Test”]
        BurnIn[“Burn-In / Stress”]
        ATE --> Probe
        Probe --> Final
        Final --> BurnIn
    end
    DFT --> ATE
    ATPG --> ATE
    BurnIn --> FA[“Failure Analysis / Qualification”]
    FA --> Ship[“Qualified Silicon Release”]
```

Figure: Test-program workflow. The design’s embedded DFT logic feeds into ATPG (pattern generation), which becomes the ATE test program. Chips are first wafer-sorted, then packaged and 100% tested functionally. Burn-in/stress and FA complete qualification.

Appendix D: ATE & Probe Card Recommendations

Parameter	Eng. Sample (Low-volume)	Pilot/Prod	Notes
ATE Platform	1× Teradyne UltraFLEX (8-slot)	2× Teradyne UltraFLEX or Advantest T2000	Multi-site needed for high throughput
Digital Channels	256 channels (e.g. 64-pin × 4 sites)	512+ channels (128-pin × 4–8 sites)	Must cover all parallel I/Os
Digital Pattern Rate	~100 MHz	300–500 MHz	Needs to handle internal clock (not necessarily CPU speed)
Analog/ Parametrics	I–V measurement units (±10V, µA resolution)	Integrated high-speed parametric PMUs	For IDDQ, V-I curves, threshold voltage tests
Power Supplies	4 programmable DC sources (0–3.3V, 0–5V)	8 or more (with low noise)	Separate rails: core, I/O, PLL, analog, etc.

Parameter	Eng. Sample (Low-volume)	Pilot/Prod	Notes
Probe Card (Wafer)	Spring-pin pogo card (~64–100 pins)	Rigid MEMS or blade card (~128–256 pins)	Layout per wafer pad grid; include high-current pogo for Vcore
Package Pogo/ Sockets (Final)	Handler sockets or single-site land-grid	Multi-site BGA sockets	Must match chosen package footprint
Required Equipment	Lab logic analyzer, scope, JTAG debugger	On-site prototyping/ test workshop	For bring-up and test development
Estimated Cost (per annum)	~$150K (ATE rental/ usage)	$300K–$500K (ATE + probe services)	Approximate for engineering/test labs

Appendix E: Failure Analysis & Escalation Matrix

Failure Symptom	Likely Cause	Analysis Method	Escalation Action
Low first-pass yield (<80%)	Systematic fault / missing DFT	Scan log review, fault dump	If repeat in >2 lots, escalate to design and yield teams
Frequent scan chain mismatches	DFT insertion error or net short	JTAG boundary scan test	Verify scan-path netlists; re-spin if needed
SRAM MBIST failures	Memory fault or decoder issue	Bit-map of failing cells, SEM	Engage memory IP vendor; adjust margins or replace comp
Unexpected parametric values	Process drift / ESD damage	Parametric tester trace, SEM	Consult fab, tighten ESD protections
Functional crash or hang	RTL bug or clock issue	JTAG debug, logic analyzer	Debug in simulation/ emulation; patch RTL or init script
Temperature-dependent failures	IR drop / thermal hotspot	Thermal imaging, power map	Improve decoupling, redistribute clocks, adjust voltage
High IDDQ (leakage)	Gate oxide defect or ESD	Microprobing, FIB, SEM	Screen out affected rows, add gate-level test
Package I/O opens/ shorts	Bonding defect	X-ray, continuity test on board	Replace package design/ vendor

Escalation: If an issue persists beyond local debug (e.g. yield <50%, critical net bug, pattern of fails), it is elevated to a cross-functional review board (design, test, fab) for corrective action (layout fix, mask repair, retest procedure). Each FA finding is documented and tied to a ticket for traceability.

Conclusion & Action Plan

In summary, SilicaFlux's test plan begins with **strong DFT and ATPG**: all flip-flops are scannable, and each SRAM block has self-test【3†L136-L144】【19†L114-L122】. IEEE-1149.1 JTAG provides final board-level access【55†L160-L167】. For silicon validation, a **reduced SilicaFlux-V prototype** will be first fabricated (as described above) to prove the flow and test programs on real hardware. Its design will include only one or two cores and minimal cache to simplify bring-up. In parallel, we will finalize the **full-chip test architecture**, develop test vectors (structural and functional), and set up the required ATE and probe equipment.

The immediate work items (in priority order) are:
1. **Complete RTL DFT Insertion:** Freeze the scan/BIST logic in the RTL and run ATPG to verify coverage.
2. **Develop and Verify Testbench:** Write ATPG scripts and preliminary test patterns; simulate them on RTL to catch obvious bugs.
3. **Plan Wafer Probe Setup:** Engage a probe card vendor, finalize pad layout, and book an MPW run for SilicaFlux-V.
4. **Design Bring-up Board:** Begin PCB layout for the sample board including connectors, regulators, and JTAG header.
5. **Establish ATE Flow:** Confirm tester availability and storage for test vectors; start writing functional tests.

Upon receiving engineering silicon, we will perform wafer-sort and bring-up steps as outlined. We will aggressively iterate test programs to isolate failures. The **release criteria** for each milestone are objective: e.g., DFT coverage $\geq$95%, wafer yield $\geq$80%, all qualification tests passed without failure. All assumptions (e.g. compression efficiency, yield rates) will be verified empirically.

References: We adhere to industry standards throughout. For example, JEDEC JESD-22 documents specify HTOL/HAST methods【39†L73-L82】 and IEEE-1149.1 defines JTAG/BScan protocols【23†L139-L148】【55†L160-L167】. These primary sources anchor our test strategy. Wherever estimates were needed (e.g. test time, costs), we have noted them as such. Further refinement will come from vendor quotations and lab measurements as we proceed.
